

MSc in Engineering – Mechatronics and Automation

Term Thesis

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NDA (Non-Disclosure Agreement)	yes
Type of work	MTh
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Subject Manufacturer-Independent Integration Architecture for Robotic Systems in Industrial Automation

Abstract

The Swiss watch industry operates in a highly competitive environment in which flexibility, traceability, and efficient production processes are essential. Robotic manipulators and stationary robotic systems offer significant potential for automating handling and production tasks, reducing non-value-adding activities, and improving the continuity and reproducibility of manufacturing operations.

However, integrating heterogeneous robotic systems into existing watch-production environments remains challenging. Relevant constraints include high requirements regarding positioning and process precision, limited workspace, interaction with production personnel, system interoperability, and industry-specific quality requirements. In addition, robotic platforms from different manufacturers typically provide different software interfaces and communication protocols, increasing engineering effort for each new automation project.

This thesis aims to develop and validate a standardized, manufacturer-independent integration concept for robotic systems within the automation projects of ETA SA. The proposed architecture will introduce an orchestration layer between the SCADA level and the robotic systems. This layer shall abstract manufacturer-specific interfaces and provide standardized communication mechanisms for different robot types.

The concept shall support robotic manipulators as well as stationary industrial and collaborative robots through a common software architecture, with the Meca500 serving as a reference system. Existing software solutions, communication mechanisms, and development standards already established within ETA SA shall be reused wherever appropriate.

The developed concept will be validated through a Proof of Concept in which a robotic manipulator performs a representative pick-and-place task. Communication between the SCADA system, the orchestration layer, and the robotic system shall be performed through the developed standardized interfaces.

The resulting architecture is intended to provide a reusable and scalable foundation for future automation projects within ETA SA and the Swatch Group, reducing integration effort while increasing flexibility in the selection and deployment of robotic platforms.

Objectives

The main objective of this thesis is to design, implement, and validate a standardized integration framework for heterogeneous robotic systems.

The project includes the following objectives:

Project registrations by email 27.07.2026

Project proposal by email 28.09.2026

Project start 15.07.2026 (required)

Report delivery* 22.1.2027 (required according to the deadlines)

Presentation date* 3.2.2027 (in general 2 weeks after report delivery)

Grade delivery* 18.2.2027 (required according to the deadlines; Advisors should communicate the provisional mark when they give feedback.)

- Analyze existing robot interfaces, APIs, and communication protocols relevant to robotic manipulators and stationary robotic systems.
- Define a standardized communication interface between the orchestration layer and the robotic systems.
- Develop a manufacturer-independent abstraction layer that separates application logic from robot-specific interfaces.
- Support robotic manipulators and stationary industrial or collaborative robots through a common architectural concept, using the Meca500 as a reference system.
- Develop a modular software architecture that enables additional robot systems and manufacturers to be integrated with limited engineering effort.
- Define a standardized workflow for integrating and commissioning new robot types.
- Reuse software technologies, communication mechanisms, and development standards already established within ETA SA wherever possible.
- Implement and validate the proposed architecture through a representative Proof of Concept.

Expected Deliverables

The expected results of the thesis are:

- A functional demonstrator consisting of a robotic manipulator and the required automation components.
- A standardized communication interface between the orchestration layer and robotic systems.
- A modular software framework for integrating additional robot systems and manufacturers.
- A documented software architecture describing the main components, responsibilities, data flows, and interfaces.
- A standardized integration and commissioning workflow for additional robot systems.
- Installation and configuration documentation for the developed system.
- A user manual including installation, configuration, commissioning, and operating instructions.
- Technical documentation of the developed software framework and its interfaces.
- A validated Proof of Concept demonstrating a manipulation task through the standardized architecture.

The proposed solution will establish a reusable and scalable software architecture, reducing engineering effort, simplifying future robot integration, and increasing flexibility in selecting robotic platforms for ETA SA and the Swatch Group.

Literature

- [1] Mecademic Inc.: User Manual for the Meca500 Industrial Robot, document MC-UM-MECA500, revision 2026.B, 2026.
- [2] Mecademic Inc.: Programming Manual for the Meca500 Industrial Robot, document MC-PM-MECA500, firmware 11.1, revision C, 2025.
- [3] OPC Foundation: OPC UA for Robotics, Part 1: Vertical Integration, OPC 40010-1, version 1.02, 2025.
- [4] ISO 10218-1:2025: Robotics - Safety requirements - Part 1: Industrial robots.

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